

Full Pipeline Report

Discovery · Inventory · Correlation · Analysis · Mitigation

PREPARED	SCOPE	COVERAGE	VERSION
2026-09-29 00:34	Workspace "ECDAT_Demo_Assets_120 · 2026-09-28 18:12"	4 scans · 11 canonical assets	ECDAT · PQC readiness

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1 Executive Summary

4 DISCOVERY SCANS	304 RAW FINDINGS	304 NORMALIZED RECORDS	11 CANONICAL ASSETS
5 (45%) SHOR-VULNERABLE	2 CLASSICALLY WEAK	38 GRAPH RELATIONS	1 COMPLETED ANALYSES
1 MITIGATION PLANS			

This assessment evaluated **4** discovery scan(s), ingesting **304** raw finding occurrences which normalized into **304** finding records and consolidated into **11** canonical cryptographic assets across **5** algorithm families.

Cryptographic Estate Breakdown:

- **5 canonical asset(s)** (45%) use classical public-key primitives (RSA/ECC/ECDSA/ECDH/DH) vulnerable to cryptographically relevant quantum computers (Shor's algorithm).
- **2 canonical asset(s)** (18%) use classically weak or deprecated primitives (e.g. MD5, SHA-1, DES) requiring immediate remediation regardless of quantum timeline.
- **4 canonical asset(s)** (36%) exhibit moderate quantum exposure (symmetric ciphers like AES-256 and SHA-2 hashes) retaining sufficient Grover/collision security margin.
- **0 canonical asset(s)** (0%) utilize approved post-quantum algorithms or quantum-resilient mechanisms.
- **0 canonical asset(s)** (0%) could not be fully classified from available repository evidence.

A mitigation plan is available (plan #14, run #14) that sequences **11** remediation action(s) across the **11** affected canonical assets: **2** action(s) in Wave 1, **5** in Wave 2, and **4** in Wave 3. **2** action(s) are URGENT priority; **5** action(s) address Shor-vulnerable public key cryptography (affecting 5 canonical assets). Estimated effort is **1.0–3.0** engineering quarter(s) (deduplicated across 11 work items).

Methodology Note: 5 canonical asset(s) (45% of canonical estate) rely on classical asymmetric primitives threatened by large-scale quantum computers. Shor vulnerability identifies theoretical quantum cryptographic exposure; remediation priority additionally considers asset criticality, network reachability, HNDL context, migration effort, business impact, and evidence confidence.

2 Scope, Threat Assumptions & Limitations

This post-quantum readiness assessment evaluates cryptographic attack surfaces across the target scope: **Workspace**
"ECDAT_Demo_Assets_120 · 2026-09-28 18:12".

Threat & Assessment Assumptions:

- **CRQC Threat Horizon (Z):** Baseline target is **2033** (7 years from 2026 assessment epoch baseline) unless overridden by organizational policy.
- **Adversary Threat Model:** Harvest-Now-Decrypt-Later (HNDL) data interception applies to long-shelf-life confidential communication and data at rest; Shor's algorithm completely breaks discrete log and factoring primitives (RSA/ECC).
- **Mosca Timeline Analysis:** Evaluated on the Mosca theorem condition: $X + Y > Z$ where X = migration complexity (yrs), Y = data shelf-life (yrs), and Z = years remaining to CRQC.

Scope Boundaries & Limitations:

- **Discovery Fidelity:** Scans inspect repository source code, AST structures, lockfiles, certificates, and infrastructure configurations. Dynamic JIT execution is observed via session telemetry.
- **Conservative Assessment:** Where operational parameters (data lifetime, network reachability, migration velocity) are missing, findings report `INSUFFICIENT_CONTEXT` / `NOT_ASSESSABLE` rather than asserting false safety.
- **Data Privacy:** Absolute developer paths and internal file system structures are sanitized and redacted.

3 Discovery Pipeline

The discovery pipeline runs borderless scans against targets, persists raw finding occurrences, normalizes and deduplicates them into normalized findings, and consolidates them into a canonical crypto asset inventory. **4** scan(s) produced **304** raw finding occurrence(s) and **304** normalized finding record(s).

ID	TARGET	SOURCE	STATUS	CREATED
99	ECDAT_Demo_Assets_120	Source Code Repos	Completed	2026-09-28 18:12:17.530960+00:00
98	ECDAT_Demo_Assets_120	Container Images	Completed	2026-09-28 18:12:17.509214+00:00
97	ECDAT_Demo_Assets_120	Certificates/PKI	Completed	2026-09-28 18:12:17.494758+00:00
96	ECDAT_Demo_Assets_120	Binary Files	Completed	2026-09-28 18:12:17.472011+00:00

Finding Occurrences by Algorithm Family

FAMILY	NORMALIZED FINDING COUNT
Hash	157
RSA	60
ECC	54
AES	20
Unknown	13

4 Inventory & Quantum Risk

11 canonical cryptographic asset(s) were consolidated from normalized findings and classified for quantum and classical security exposure: **5 Shor-vulnerable** (asymmetric public key), **2 classically weak** (MD5/SHA-1/DES), **4 moderate** (symmetric ciphers / secure hashes), **0 PQC-ready**, and **0 unknown / insufficient context**.

ASSET	FAMILY	ALGORITHM	KEY / PARAMETER SIZE	CURVE	RISK CLASSIFICATION	ROLE-AWARE RECOMMENDATION
workspace - MD5	Hash	MD5	—	—	Weak (Classical)	SHA-256 / SHA-3
workspace - SHA-1	Hash	SHA-1	—	—	Weak (Classical)	SHA-256 / SHA-3
CN=ecdat-demo.local,O=ECDAT Demo,L=Demo,ST=Demo,C=IN [RSA]	RSA	RSA	2048	—	Vulnerable (Shor)	ML-KEM-768 (Key Exchange) / ML-DSA-65 (Signature)
workspace - ECC	ECC	ECC	—	—	Vulnerable (Shor)	ML-KEM-768 (Key Exchange) / ML-DSA-65 (Signature)
workspace - Ed25519	ECC	Ed25519	256 (Curve25519)	—	Vulnerable (Shor)	ML-DSA-65 (FIPS 204) / SLH-DSA
workspace - RSA	RSA	RSA	2048	—	Vulnerable (Shor)	ML-KEM-768 (Key Exchange) / ML-DSA-65 (Signature)
workspace - RSA	RSA	RSA	2048	—	Vulnerable (Shor)	ML-KEM-768 (Key Exchange) / ML-DSA-65 (Signature)
workspace - AES	AES	AES	256	—	Moderate (Grover Margin)	AES-256-GCM - Retain (Strong)
workspace - SHA-256	Hash	SHA-256	—	—	Moderate (Grover Margin)	SHA-256 / SHA-3 - Retain (Strong)
workspace - SHA-384	Hash	SHA-384	—	—	Moderate (Grover Margin)	SHA-256 / SHA-3 - Retain (Strong)
workspace - SHA-512	Hash	SHA-512	—	—	Moderate (Grover Margin)	SHA-256 / SHA-3 - Retain (Strong)

Migration Standards: Public-key key exchange is transitioned to ML-KEM-768/1024 (FIPS 203) or hybrid X25519MLKEM768; digital signatures transition to ML-DSA-65/87 (FIPS 204) or SLH-DSA (FIPS 205); symmetric encryption retains AES-256-GCM; legacy hashes (MD5/SHA-1) migrate to SHA-256 / SHA-3.

5 Analysis & Assessment

Each completed run executes the full ECDAT assessment pipeline — CBOM consolidation, multi-dimensional risk classification, evidence-driven HNDL (Harvest-Now-Decrypt-Later) modeling, and MOSCA (Migration of Post-Quantum Cryptography) timeline scoring.

Run #14 · ECDAT_Demo_Assets_120

STATUS

Completed

FINDING RECORDS ASSESSED

304

URGENT PRIORITY

114

CRITICAL RISK

182

HNDL-APPLICABLE

147

CREATED

2026-09-28 18:12:38.967741+00:00

Mosca Threat Parameters:

Migration Duration (X): 3 yrs · Data Shelf-Life (Y): 8 yrs · CRQC Horizon: 2033 (Z = 7 yrs until CRQC) · Network Reachability: Internet-Facing · Mosca Condition: X + Y > Z (At Risk / Urgent)

ASSET	ALGORITHM	CATEGORY	CLASSICAL SEC	HNDL EXPOSURE	OVERALL RISK	PRIORITY	SHOR VULNERABLE
N5246	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5245	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5244	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5243	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5242	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5241	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5240	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5239	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5238	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5237	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5236	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5235	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5234	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5233	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5232	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5231	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5230	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5229	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5228	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5227	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5226	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5225	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5224	SHA-1	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5223	SHA-1	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5222	SHA-1	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5221	SHA-1	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5220	SHA-1	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5219	SHA-1	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5218	SHA-1	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5217	SHA-1	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5216	SHA-1	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5215	SHA-1	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5214	SHA-1	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No

ASSET	ALGORITHM	CATEGORY	CLASSICAL SEC	HNDL EXPOSURE	OVERALL RISK	PRIORITY	SHOR VULNERABLE
N5213	SHA-1	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5212	SHA-1	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5211	SHA-256	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5210	SHA-256	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5209	SHA-256	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5208	SHA-256	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5207	SHA-256	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5206	SHA-256	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5205	SHA-256	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5204	SHA-384	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5203	SHA-384	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5202	SHA-384	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5201	SHA-384	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5200	SHA-512	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5199	SHA-512	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5198	SHA-512	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5197	SHA-512	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5196	SHA-512	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5195	Ed25519	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5194	Ed25519	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5193	Ed25519	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5192	Ed25519	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5191	Ed25519	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5190	Ed25519	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5189	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5188	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5187	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5186	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5185	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5184	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5183	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5182	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5181	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5180	Ed25519	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5179	SHA-1	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5178	SHA-1	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5177	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5176	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5175	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5174	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5173	AES	SYMMETRIC	—	LOW	LOW	LOW	No
N5172	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No

ASSET	ALGORITHM	CATEGORY	CLASSICAL SEC	HNDL EXPOSURE	OVERALL RISK	PRIORITY	SHOR VULNERABLE
N5171	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5170	TLS	UNKNOWN	—	NOT_ASSESSABLE	NEGLIGIBLE	NEGLIGIBLE	No
N5169	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5168	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5167	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5166	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5165	TLS	UNKNOWN	—	NOT_ASSESSABLE	NEGLIGIBLE	NEGLIGIBLE	No
N5164	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5163	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5162	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5161	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5160	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5159	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5158	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5157	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5156	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5155	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5154	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5153	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5152	TLS	UNKNOWN	—	NOT_ASSESSABLE	NEGLIGIBLE	NEGLIGIBLE	No
N5151	tls	UNKNOWN	—	NOT_ASSESSABLE	NEGLIGIBLE	NEGLIGIBLE	No
N5150	tls	UNKNOWN	—	NOT_ASSESSABLE	NEGLIGIBLE	NEGLIGIBLE	No
N5149	TLS	UNKNOWN	—	NOT_ASSESSABLE	NEGLIGIBLE	NEGLIGIBLE	No
N5148	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5147	TLS	UNKNOWN	—	NOT_ASSESSABLE	NEGLIGIBLE	NEGLIGIBLE	No
N5146	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5145	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5144	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5143	TLS	UNKNOWN	—	NOT_ASSESSABLE	NEGLIGIBLE	NEGLIGIBLE	No
N5142	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5141	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5140	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5139	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5138	tls	UNKNOWN	—	NOT_ASSESSABLE	NEGLIGIBLE	NEGLIGIBLE	No
N5137	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5136	TLS	UNKNOWN	—	NOT_ASSESSABLE	NEGLIGIBLE	NEGLIGIBLE	No
N5135	tls	UNKNOWN	—	NOT_ASSESSABLE	NEGLIGIBLE	NEGLIGIBLE	No
N5134	TLS	UNKNOWN	—	NOT_ASSESSABLE	NEGLIGIBLE	NEGLIGIBLE	No
N5133	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5132	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5131	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5130	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No

ASSET	ALGORITHM	CATEGORY	CLASSICAL SEC	HNDL EXPOSURE	OVERALL RISK	PRIORITY	SHOR VULNERABLE
N5129	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5128	SHA-1	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5127	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5126	SHA-384	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5125	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5124	SHA-512	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5123	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5122	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5121	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5120	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5119	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5118	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5117	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5116	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5115	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5114	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5113	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5112	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5111	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5110	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5109	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5108	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5107	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5106	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5105	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5104	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5103	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5102	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5101	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5100	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5099	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5098	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5097	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5096	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5095	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N5094	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5093	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5092	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5091	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5090	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5089	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5088	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)

ASSET	ALGORITHM	CATEGORY	CLASSICAL SEC	HNDL EXPOSURE	OVERALL RISK	PRIORITY	SHOR VULNERABLE
N5087	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5086	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5085	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5084	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5083	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5082	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5081	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5080	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5079	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5078	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5077	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5076	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5075	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5074	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5073	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5072	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5071	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5070	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5069	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5068	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N5067	AES	SYMMETRIC	—	LOW	LOW	LOW	No
N5066	AES	SYMMETRIC	—	LOW	LOW	LOW	No
N5065	AES	SYMMETRIC	—	LOW	LOW	LOW	No
N5064	AES	SYMMETRIC	—	LOW	LOW	LOW	No
N5063	AES	SYMMETRIC	—	LOW	LOW	LOW	No
N5062	AES	SYMMETRIC	—	LOW	LOW	LOW	No
N5061	AES	SYMMETRIC	—	LOW	LOW	LOW	No
N5060	AES	SYMMETRIC	—	LOW	LOW	LOW	No
N5059	AES	SYMMETRIC	—	LOW	LOW	LOW	No
N5058	AES	SYMMETRIC	—	LOW	LOW	LOW	No
N5057	AES	SYMMETRIC	—	LOW	LOW	LOW	No
N5056	AES	SYMMETRIC	—	LOW	LOW	LOW	No
N5055	AES	SYMMETRIC	—	LOW	LOW	LOW	No
N5054	AES	SYMMETRIC	—	LOW	LOW	LOW	No
N5053	AES	SYMMETRIC	—	LOW	LOW	LOW	No
N5052	AES	SYMMETRIC	—	LOW	LOW	LOW	No
N5051	AES	SYMMETRIC	—	LOW	LOW	LOW	No
N5050	AES	SYMMETRIC	—	LOW	LOW	LOW	No
N5049	AES	SYMMETRIC	—	LOW	LOW	LOW	No
N5048	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5047	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5046	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No

ASSET	ALGORITHM	CATEGORY	CLASSICAL SEC	HNDL EXPOSURE	OVERALL RISK	PRIORITY	SHOR VULNERABLE
N5045	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5044	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5043	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5042	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5041	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5040	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5039	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5038	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5037	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5036	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5035	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5034	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5033	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5032	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5031	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5030	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5029	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5028	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5027	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5026	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5025	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5024	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5023	SHA-1	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5022	SHA-1	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5021	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5020	SHA-1	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5019	SHA-1	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5018	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5017	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5016	SHA-1	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5015	SHA-1	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5014	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5013	SHA-1	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5012	SHA-1	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5011	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5010	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5009	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5008	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5007	SHA-1	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5006	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5005	SHA-1	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5004	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No

ASSET	ALGORITHM	CATEGORY	CLASSICAL SEC	HNDL EXPOSURE	OVERALL RISK	PRIORITY	SHOR VULNERABLE
N5003	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5002	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N5001	SHA-1	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N5000	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4999	SHA-1	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N4998	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4997	SHA-256	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4996	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4995	SHA-256	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4994	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4993	SHA-256	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4992	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4991	SHA-256	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4990	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4989	SHA-256	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4988	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4987	SHA-256	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4986	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4985	SHA-256	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4984	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4983	SHA-384	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4982	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4981	SHA-384	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4980	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4979	SHA-384	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4978	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4977	SHA-384	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4976	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4975	SHA-512	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4974	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4973	SHA-512	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4972	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4971	SHA-512	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4970	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4969	SHA-512	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4968	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4967	SHA-512	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4966	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4965	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N4964	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N4963	Ed25519	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N4962	Ed25519	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)

ASSET	ALGORITHM	CATEGORY	CLASSICAL SEC	HNDL EXPOSURE	OVERALL RISK	PRIORITY	SHOR VULNERABLE
N4961	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N4960	Ed25519	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N4959	Ed25519	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N4958	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N4957	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N4956	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N4955	Ed25519	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N4954	Ed25519	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N4953	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N4952	Ed25519	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N4951	Ed25519	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N4950	ECC	PUBLIC_KEY	—	LOW	HIGH	HIGH	Yes (Shor)
N4949	MD5	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N4948	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4947	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N4946	SHA-1	HASH	WEAK	NEGLIGIBLE	HIGH	HIGH	No
N4945	HASH	HASH	—	NEGLIGIBLE	LOW	LOW	No
N4944	RSA	PUBLIC_KEY	—	MEDIUM	HIGH	HIGH	Yes (Shor)
N4943	KEY	UNKNOWN	—	NOT_ASSESSABLE	NEGLIGIBLE	NEGLIGIBLE	No

6 Mitigation Plan

PLAN	ANALYSIS RUN	GENERATED	NARRATION	WAVE ENGINE
#14	#14	2026-09-28 19:04:07.842583+00:00	AI-enhanced · gemini-3.1-flash-lite	deterministic rule engine
11 REMIATION ACTIONS (11 CANONICAL ASSETS) · W1 2 W2 5 W3 4	2 / 7 URGENT / HIGH-RISK ACTIONS	5 SHOR-VULN ACTIONS	1.0–3.0 qtrs EFFORT RANGE (QUARTERS)	

Executive summary

The ECDAT_Demo_Assets_120 application contains 11 cryptographic assets, with 5 identified as quantum-vulnerable and 7 classified as high-risk. Remediation is structured into three waves, prioritizing the mitigation of classical-weak primitives and the transition to post-quantum standards. The total effort is estimated between 1 and 3 quarters, focusing on replacing vulnerable public-key primitives while retaining quantum-resilient symmetric and hash functions. This plan ensures long-term security against both current threats and future quantum-enabled decryption.

Quantum Risk: Public-key primitives like RSA and ECC are susceptible to Shor's algorithm, which threatens the confidentiality of captured data through Harvest-Now-Decrypt-Later (HNDL) attacks. Migrating to quantum-resistant algorithms is essential to protect long-lived data and ensure the integrity of digital signatures against future cryptographically-relevant quantum computers.

Shor-Vulnerable Assets: RSA · asset-410 · certs RSA · asset-412 · tests ECC · asset-414 · service_112_ed25519.java ED25519 · asset-415 · service_112_ed25519.java RSA · asset-420 · service_013_rsa.py

Digital twin

Context: app ECDAT_Demo_Assets_120 · exposure internal

Data types: none labelled · **Services:** 8

ABSTRACTION	ROLES	ASSETS	CONTEXTS
RSA	digital_signature, authentication, certificates	3	certs, tests, service_013_rsa.py
SHA-1	integrity, message_digest	1	tests
MD5	integrity, message_digest	1	tests
ECC	cryptographic_primitives	1	service_112_ed25519.java
ED25519	digital_signature, authentication, certificates	1	service_112_ed25519.java
SHA-512	integrity, message_digest	1	service_106_sha512.js
SHA-384	integrity, message_digest	1	service_101_sha384.py
SHA-256	integrity, message_digest	1	service_097_sha256.py
AES	confidentiality, data_encryption	1	service_048_aes.js

High-value points

- **Service 'tests'** — concentrates 3 cryptographic asset(s) in one blast zone
- **Service 'service_112_ed25519.java'** — concentrates 2 cryptographic asset(s) in one blast zone
- **Service 'certs'** — concentrates 1 cryptographic asset(s) in one blast zone
- **Service 'service_106_sha512.js'** — concentrates 1 cryptographic asset(s) in one blast zone

Blast radius

Worst exposure across 8 service(s) and 11 canonical remediation work item(s) — internal-only. Severity: CRITICAL

ALGORITHM	ASSETS	SEVERITY
RSA	3	CRITICAL
SHA-1	1	CRITICAL
ECC	1	CRITICAL
ED25519	1	CRITICAL
MD5	1	MEDIUM
SHA-512	1	LOW
SHA-384	1	LOW

ALGORITHM	ASSETS	SEVERITY
SHA-256	1	LOW
AES	1	LOW

Poorest link: CRITICAL — the weakest primitive bounds the estate's overall quantum readiness.

Remediation Waves

WAVE 1 · 0-3 MONTHS Wave 1 — Classical-weak remediation & HNDL triage Remediate classically broken/deprecated primitives, address HNDL-exposed confidentiality paths through algorithm migration or hybridization (rotating keys/certificates only where key hygiene, expiry, weakness, or compromise requires rotation), add hybrid handshakes, and stage urgent PQC replacements. workspace — SHA-1 workspace — MD5	WAVE 2 · 3-12 MONTHS Wave 2 — Post-quantum migration Systematically move public-key signatures and key exchange to approved post-quantum mechanisms (ML-KEM / ML-DSA), covering downstream trust chains. CN=ecdat-demo.local,O=ECDAT Demo,L=Demo,ST=Demo,C=IN [RSA] workspace — RSA workspace — ECC workspace — Ed25519 workspace — RSA	WAVE 3 · 12-24 MONTHS Wave 3 — Symmetric hardening & governance Consolidate symmetric modes (AES-256-GCM), strengthen hashes to SHA-256+/SHA-3, retire legacy ciphers, and codify crypto policy so new failures cannot ship. workspace — SHA-512 workspace — SHA-384 workspace — SHA-256 workspace — AES
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Strategic recommendations

RECOMMENDATION Wave 1: Classical-Weak Remediation Address legacy cryptographic primitives that are broken classically, such as outdated hash functions or weak symmetric ciphers, to reduce immediate attack surface. Timeline: 0-3 months	RECOMMENDATION Wave 2: Post-Quantum Migration Execute the migration of RSA and ECC-based public-key infrastructure to NIST-approved FIPS 203 (ML-KEM) and FIPS 204 (ML-DSA) algorithms. Timeline: 3-12 months	RECOMMENDATION Wave 3: Symmetric Hardening & Governance Finalize the transition to AES-256-GCM and SHA-3, while establishing long-term cryptographic governance and key rotation policies. Timeline: 12-24 months
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Prioritized Remediation Actions

ASSET	ALGORITHM	PRIORITY	BLAST	ROLE-AWARE REPLACEMENT	EFFORT	WAVE	REMEDIACTION GUIDANCE
workspace — SHA-1	SHA-1	URGENT	CRITICAL 3 shared · tests	SHA-256 / SHA-3	LOW	1	- Migrate deprecated SHA-1 in 'tests' to approved hash construction (SHA-256 / SHA-3). - Remove SHA-1 from security-sensitive signature and digest paths. - Treat as a Week-0 item: raise a change ticket and assign an owner.
workspace — MD5	MD5	URGENT	MEDIUM 3 shared · tests	Argon2id / PBKDF2-HMAC-SHA256	LOW	1	- Migrate deprecated MD5 in 'tests' to approved hash construction (Argon2id / PBKDF2-HMAC-SHA256). - Remove MD5 from security-sensitive signature and digest paths. - Treat as a Week-0 item: raise a change ticket and assign an owner.
CN=ecdat-demo.local,O=ECDAT Demo,L=Demo,ST=Demo,C=IN [RSA]	RSA Short-Vuln	HIGH	CRITICAL 1 shared · certs	ML-DSA-87 (FIPS 204)	HIGH	2	- Migrate RSA to ML-DSA-87 (FIPS 204) (NIST General (FIPS 203/204/205)) in 'certs'. - Consider hybrid key establishment using X25519MLKEM768 where protocol support and interoperability requirements permit. - Classify data protected

ASSET	ALGORITHM	PRIORITY	BLAST	ROLE-AWARE REPLACEMENT	EFFORT	WAVE	REMEDIATION GUIDANCE
							by RSA and sequence algorithm migration to an approved PQC scheme. • Schedule within the current migration quarter; assign an owner.
workspace – RSA	RSA Short-Vuln	HIGH	CRITICAL 3 shared · tests	ML-KEM-1024 / ML-DSA-87	HIGH	2	• Migrate RSA to ML-KEM-1024 / ML-DSA-87 (NIST General (FIPS 203/204/205)) in 'tests'. • Consider hybrid key establishment using X25519MLKEM768 where protocol support and interoperability requirements permit. • Classify data protected by RSA and sequence algorithm migration to an approved PQC scheme. • Schedule within the current migration quarter; assign an owner.
workspace – ECC	ECC Short-Vuln	HIGH	CRITICAL 2 shared · service_112_ed25519.java	ML-KEM-1024 (Key Exchange) / ML-DSA-65 (Signatures)	MEDIUM	2	• Migrate ECC to ML-KEM-1024 (Key Exchange) / ML-DSA-65 (Signatures) (NIST General (FIPS 203/204/205)) in 'service_112_ed25519.java'. • Consider hybrid key establishment using X25519MLKEM768 where protocol support and interoperability requirements permit. • Classify data protected by ECC and sequence algorithm migration to an approved PQC scheme. • Schedule within the current migration quarter; assign an owner.
workspace – Ed25519	ED25519 Short-Vuln	HIGH	CRITICAL 2 shared · service_112_ed25519.java	ML-DSA-65 (FIPS 204)	MEDIUM	2	• Migrate ED25519 to ML-DSA-65 (FIPS 204) (NIST General (FIPS 203/204/205)) in 'service_112_ed25519.java'. • Classify data protected by ED25519 and sequence algorithm migration to an approved PQC scheme. • Schedule within the current migration quarter; assign an owner.
workspace – RSA	RSA Short-Vuln	HIGH	CRITICAL 1 shared · service_013_rsa.py	ML-KEM-1024 / ML-DSA-87	HIGH	2	• Migrate RSA to ML-KEM-1024 / ML-DSA-87 (NIST General (FIPS 203/204/205)) in 'service_013_rsa.py'. • Consider hybrid key establishment using X25519MLKEM768 where protocol support and interoperability requirements permit. • Classify data protected by RSA and sequence algorithm migration to an approved PQC scheme. • Schedule within the current migration quarter; assign an owner.

ASSET	ALGORITHM	PRIORITY	BLAST	ROLE-AWARE REPLACEMENT	EFFORT	WAVE	REMEDIATION GUIDANCE
workspace – SHA-512	SHA-512	LOW	LOW 1 shared · service_106_sha512.js	SHA-512 / SHA-3 – Retain (Strong)	LOW	3	• Retain SHA-512 for integrity and digest contexts in 'service_106_sha512.js'. • Maintain routine cryptographic hygiene and key rotation for HMAC constructions.
workspace – SHA-384	SHA-384	LOW	LOW 1 shared · service_101_sha384.py	SHA-384 / SHA-3 – Retain (Strong)	LOW	3	• Retain SHA-384 for integrity and digest contexts in 'service_101_sha384.py'. • Maintain routine cryptographic hygiene and key rotation for HMAC constructions.
workspace – SHA-256	SHA-256	LOW	LOW 1 shared · service_097_sha256.py	SHA-256 / SHA-3 – Retain (Strong)	LOW	3	• Retain SHA-256 for integrity and digest contexts in 'service_097_sha256.py'. • Maintain routine cryptographic hygiene and key rotation for HMAC constructions.
workspace – AES	AES	LOW	LOW 1 shared · service_048_aes.js	AES-256 (GCM) – Retain (Strong)	LOW	3	• Retain AES with approved AEAD mode (e.g. GCM); validate key-management and rotation in 'service_048_aes.js'. • Ensure symmetric keys are rotated on schedule and stored in a KMS.

7 Analysis Invariants & Integrity Audit

Formal semantic integrity verification executed automatically over the analysis result: **[ALL INVARIANTS SATISFIED]**

INVARIANT CHECK	STATUS	AUDIT DETAILS & VALIDATION EVIDENCE
INV-01: Mosca Threat Propagation	PASS	Risk priority and HNDL tiers correctly reflect Mosca threat parameters.
INV-02: Classical Weakness Priority Floor	PASS	All classically weak primitives (MD5, SHA-1, DES) receive elevated remediation priority.
INV-03: Summary vs Table Consistency	PASS	Executive summary canonical count (11) exactly matches inventory table rows (11).
INV-04: Bare Keyword Isolation	PASS	No bare keywords (crypto, KEY, TLS, HASH) exist as standalone canonical assets.
INV-05: Key-Size Precision	PASS	Non-key primitives (hashes, digests, protocols) carry no synthetic key sizes.
INV-06: Cryptographic Role Separation	PASS	All Shor-vulnerable public key assets have unambiguous role assignments.
INV-07: Dynamic Wave Sequencing	PASS	Remediation waves dynamically sequence Shor-vulnerable migrations.
INV-08: Transparent Narration Provenance	PASS	Narration labeling matches engine mode (AI-Enhanced with gemini-3.1-flash-lite).
INV-09: Single Source of Truth	PASS	Single canonical asset collection (11 canonical assets) uniformly referenced across all pipeline stages.
INV-10: Mitigation Plan Integrity & Work-Item Consistency	PASS	Mitigation plan staged 11 canonical work item(s) across 3 waves (W1: 2, W2: 5, W3: 4) with 1.0–3.0 qtrs effort range.

8 Appendix

Method notes

- **CBOM** — Cryptographic Bill of Materials: every crypto artefact (algorithms, keys, certificates, protocols, libraries, HSMS, cloud services) discovered across the estate.
- **HNDL** — Harvest-Now-Decrypt-Later: contextual threat modeling assessing whether encrypted data in transit or storage can be captured today and decrypted when CRQCs emerge. When required operational parameters (data shelf-life, sensitivity, network exposure) are absent, status is marked NOT_ASSESSABLE rather than claiming false safety.
- **MOSCA** — Migration of Post-Quantum Cryptography scoring: $X + Y > Z$, where X = migration duration (years), Y = data shelf-life (years), and Z = years remaining until CRQC threat horizon (e.g., $2033 - 2026 = 7$ years). If $X + Y > Z$, migration will not complete before quantum threat emergence, creating a migration deficit.
- **Data Hierarchy & Semantic Terminology** — Raw findings represent individual scanner syntax matches; normalized findings consolidate syntactic duplicates; canonical assets represent unique consolidated cryptographic attack surfaces. Remediation actions count actionable tasks sequenced across the affected assets.

Run ledger

Analysis run IDs in scope, newest first: 14. Graph correlation produces asset-to-asset relations within the estate (38 edge(s)).